

Multi-Agent Receding-Horizon Games for Autonomous Market Participation

Detailed Case-Study Report — Distributed Day-Ahead ADMM & Explicit
Real-Time FACET Generalized Nash Equilibrium
Green-H₂ coalition, ERCOT 2025, $H = 4$ (1-hour) receding horizon, PCC band [100, 900] kW

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Abstract

Six independently-owned green-hydrogen electrolyzers behind one point-of-common-coupling (PCC) bid collectively into the ERCOT two-settlement market. **Both** market stages are solved fully distributed, over the *same* physical PCC import band $[L_{\min}, L_{\max}] = [100, 900]$ kW, and *no agent ever shares its cost model*. The day-ahead (DAM) anchor is set by a distributed **ADMM** in which each agent solves only its own 24-hour QP and the sole shared quantity is the per-hour coalition import. Real-time dispatch (RTM, 15-min) is a **receding-horizon** ($H = 4$) parametric generalized Nash equilibrium (GNE) solved online by **FACET**: each agent locates its own precomputed critical region by warm-started point-location over its facet-neighbour graph

(never enumerating the $K^N \approx 9.7 \times 10^{20}$ combinations), the coalition combination is made self-consistent, and the *variational* GNE (= social optimum) is recovered by a potential-minimizing solve that remains valid when the coupling binds. Over two independent ERCOT weeks (1–7 April, 7–13 July 2025; $-\$1$ to $\$3553/\text{MWh}$) FACET clears **99.1%** of steps *iteration-free* — exactly one decision broadcast per agent per step — meeting the weekly H_2 target at **140–142%**, holding renewable curtailment to **6.9–7.8%** of available generation, and matching the centralized equilibrium to $\sim 10^{-4}$ kW in the interior. On the 0.9% of steps at the degenerate coupling ceiling a warm-started distributed ADMM fallback is used (the only iteration). This report documents each agent’s optimization problem, the distributed DAM and the online FACET clearing step by step, and all measured results including curtailment.

1 Physical system and market

A cluster of small electrolyzers sits behind one shared PCC. Each owner runs its own asset for its own revenue and keeps its cost model private; only power decisions are shared. Two forces make this a game rather than N independent problems:

- **Physical coupling.** The PCC caps the collective import and a market floor sets a minimum offer, *at every 15-min step k* :

$$L_{\min} = 100 \text{ kW} \leq \sum_{i=1}^N p_{i,k} \leq L_{\max} = 900 \text{ kW}.$$

$L_{\min}$ is the ISO minimum offer size; $L_{\max}$ is the physical interconnection limit (75% of the 1200 kW fleet nameplate). **This same band governs both the day-ahead and the real-time stage** — there is one physical wire.

- **Private hydrogen contracts.** Each agent must produce a daily H_2 quantity; it may time production freely but the daily total is a hard floor. That freedom, under a shared per-step power limit, is what couples the agents and the horizon.

The market is ERCOT two-settlement: a day-ahead (DAM, hourly) stage that sets a financial anchor p^{DA} , and a real-time (RTM, 15-min SCED) stage that re-optimizes against live prices. Electricity $\rightarrow \text{H}_2$ conversion: $\text{kg} = \eta_i p_{\text{elec},i} \Delta t$, $\Delta t = 0.25 \text{ h}$.

2 The fleet (medium-scale)

i	Agent	type	$P_{\max,i}$ [kW]	renew. cap [kW]	η_i [kg/kWh]	γ_i	break-even a_i [\$/MWh]
0	PEM_Elec	grid	250	–	0.020	5×10^{-3}	60
1	ALK	grid	200	–	0.018	5×10^{-3}	54
2	PEM_PV	PV	125	125	0.020	5×10^{-3}	60
3	PEM_PV_2	PV	125	125	0.020	5×10^{-3}	60
4	PEM_Wind	wind	250	250	0.020	4×10^{-3}	60
5	PEM_Wind_2	wind	250	250	0.020	4×10^{-3}	60

Table 1: $N = 6$, total nameplate 1200 kW. $r_{\text{H}_2} = \$3/\text{kg}$, $H = 4$, $\Delta t = 0.25 \text{ h}$. Break-even $a_i = r_{\text{H}_2} \eta_i \cdot 10^3$: below a_i buying grid power to make H_2 is profitable, above it the agent wants to stop. Agents 2–3 and 4–5 are identical pairs (they share one offline solve each).

3 Day-Ahead stage: distributed ADMM (model-private)

The day-ahead schedule is solved once per day on the DAM *forecast* ($\hat{\lambda}_h = \lambda_h^{\text{DA}} + \mathcal{N}(0, 16)$), renewable $\hat{g}_{i,h}$. It is **distributed**: each agent i solves only its own 24-hour problem, and the *only* exchanged quantity is the per-hour coalition import $\sum_i p_{i,h}$ (already public at the metered point). No agent reveals $P_{\max,i}$, η_i , its renewable cap, or its daily target.

Per-agent day-ahead problem. With $\Delta t_{\text{DA}} = 1$ h and daily target $D_i^{\text{day}} = 0.55 P_{\max,i} \eta_i \cdot 24$, agent i solves

$$\min_{p_{i,h} \geq 0, cv_{i,h} \geq 0} \sum_{h=0}^{23} \left[(\hat{\lambda}_h/1000 - r_{\text{H2}} \eta_i) \Delta t_{\text{DA}} p_{i,h} + r_{\text{H2}} \eta_i \Delta t_{\text{DA}} cv_{i,h} + \underbrace{\frac{1}{2} \gamma_{\text{DA}} p_{i,h}^2}_{\text{regularizer}} \right]$$

$$0 \leq p_{i,h} \leq P_{\max,i}, \quad 0 \leq cv_{i,h} \leq \text{cap}_i \quad (\text{local box; } cv \text{ only renewables})$$

$$0 \leq p_{i,h} + \hat{g}_{i,h} - cv_{i,h} \leq P_{\max,i} \quad (\text{local renewable balance})$$

$$\sum_{h=0}^{23} \eta_i (p_{i,h} + \hat{g}_{i,h} - cv_{i,h}) \Delta t_{\text{DA}} \geq D_i^{\text{day}} \quad (\text{hard daily H}_2 \text{ floor})$$

Every term uses only agent i 's own data; the daily H₂ floor couples that agent's own 24 hours. The **coupling band** $L_{\min} \leq \sum_i p_{i,h} \leq L_{\max}$ ($\forall h$) is the *only* cross-agent constraint, enforced by ADMM consensus/dual updates that see nothing but the per-hour aggregate:

$$(x) p_i^{\ell+1} = \arg \min [J_i(p_i) + \frac{\rho}{2} \|p_i - z_i^\ell\|^2], \quad (z) z^{\ell+1} = \Pi_{[L_{\min}, L_{\max}]}(\sum_i p_i^{\ell+1} + \mu^\ell / \rho), \quad (\mu) \mu_i^{\ell+1} = \mu_i^\ell + \rho(p_i^{\ell+1} - z_i^{\ell+1}),$$

with $\rho = 0.1$; Π is a closed-form clip of the per-hour aggregate onto the band. The award $p_{i,h}^{\text{DA}}$ is the RTM anchor; its cumulative-H₂ trajectory sets the RTM pacing target $D_i(t)$. (`dam.solve_dam_admm`; consensus in `admm_solver.py`.)

The small γ_{DA} regularizer (why, and its trade-off)

The day-ahead economics are *linear* in each hourly buy: “buy enough cheap power to make the day's hydrogen.” With no γ_{DA} the problem is a degenerate LP — on a cheap day it says *buy the maximum every hour*, a knife-edge answer that ADMM oscillates on and cannot settle. A *tiny* quadratic penalty $\frac{1}{2} \gamma_{\text{DA}} p_{i,h}^2$ ($\gamma_{\text{DA}} = 10^{-4}$) rounds the knife-edge: it spreads purchases smoothly, gives a *unique* schedule so ADMM converges, and changes total cost by $\leq 1.4\%$. A matching $\varepsilon = 10^{-3}$ regularizes renewable curtailment cv . Validation: a centralized solve of the *same* regularized problem (offline oracle, `solve_dam_centralized`) is matched to ≤ 10 kW on priced days; on ultra-flat days the day-ahead problem is genuinely degenerate so the two solvers land on different equally-optimal schedules (≤ 23 kW apart, cost within 1%).

4 Real-Time stage: each agent's mpQP

Every RTM step, agent i solves a strictly-convex mpQP over the horizon $k = 0, \dots, 3$. The decision is grid import $p_{i,k} \geq 0$ (renewable agents additionally choose curtailment $cv_{i,k}$). The cost is a day-ahead deviation anchor $\frac{1}{2} \gamma_i (p - p_i^{\text{DA}})^2$ (strict convexity, $Q_i \succ 0$) plus real-time energy cost net of H₂ revenue. **Each agent's cost and every local constraint use only that agent's own parameters; the sole shared quantity is the aggregate** $s_k = \sum_{j \neq i} p_{j,k}$.

4.1 Grid agents (PEM_Elec $i=0$, ALK $i=1$)

Grid agent i — decision $x_i = [p_{i,0..3}] \in \mathbb{R}^4$

Private parameters $\theta_i = [s_0..s_3 \mid D_i, \lambda_0..\lambda_3, p_i^{\text{DA}}] \in \mathbb{R}^{10}$

$$\begin{aligned} \min_{p_i \in \mathbb{R}^4} \quad & \sum_{k=0}^3 \left[\frac{1}{2} \gamma_i (p_{i,k} - p_i^{\text{DA}})^2 + (\lambda_k/1000 - r_{\text{H}_2} \eta_i) \Delta t p_{i,k} \right] \\ 0 \leq p_{i,k} \leq P_{\max,i}, \quad & \text{(capacity box)} \\ \sum_k \eta_i p_{i,k} \Delta t \geq D_i, \quad & \text{(cumulative H}_2 \text{ — couples the 4 steps)} \\ L_{\min} \leq p_{i,k} + s_k \leq L_{\max}. \quad & \text{(shared per-step coupling)} \end{aligned}$$

4.2 Renewable agents (PEM_PV $i=2, 3$; PEM_Wind $i=4, 5$)

Renewable output $g_{i,k}$ feeds the electrolyzer load $p_{\text{elec},i,k} = p_{i,k} + g_{i,k} - cv_{i,k}$ (free renewable displaces grid import; curtailment cv throws away H_2 at break-even).

Renewable agent i — decision $x_i = [p_{i,0..3}, cv_{i,0..3}] \in \mathbb{R}^8$

Private parameters $\theta_i = [s_0..s_3 \mid D_i, \lambda_0..\lambda_3, p_i^{\text{DA}}, g_0..g_3] \in \mathbb{R}^{14}$

$$\begin{aligned} \min_{p_i, cv_i} \quad & \sum_{k=0}^3 \left[\frac{1}{2} \gamma_i (p_{i,k} - p_i^{\text{DA}})^2 + \frac{1}{2} \varepsilon cv_{i,k}^2 + (\lambda_k/1000 - r_{\text{H}_2} \eta_i) \Delta t p_{i,k} + r_{\text{H}_2} \eta_i \Delta t cv_{i,k} \right] \\ 0 \leq p_{i,k} \leq P_{\max,i}, \quad 0 \leq cv_{i,k} \leq g_{i,k} \leq \text{cap}_i, \\ 0 \leq p_{\text{elec},i,k} = p_{i,k} + g_{i,k} - cv_{i,k} \leq P_{\max,i}, \\ \sum_k \eta_i p_{\text{elec},i,k} \Delta t \geq D_i, \quad L_{\min} \leq p_{i,k} + s_k \leq L_{\max} \text{ (grid import only)}. \end{aligned}$$

Receding-horizon coupling (not a repeated static game). The four steps couple only through the cumulative- H_2 row. At an H_2 -binding point (PEM, raise λ_1 by \$40/MWh): $\Delta p = [+0.5, -1.5, +0.5, +0.5]$ kW with $\sum_k p_k : 600 \rightarrow 600$ (H_2 -conserved). Step 1 becomes expensive so it drops; the others rise to keep the H_2 total — a single-step map would give $\Delta p = \mathbf{0}$.

Sum-compression and elimination. Agent i depends on the others only through the per-step aggregate $s_k = \sum_{j \neq i} p_{j,k}$ (4 numbers), not their private $D_j, p_j^{\text{DA}}, g_j$. Each agent's mpQP is solved offline (PPOPT) in its private 10- or 14-dimensional θ_i ; the coupling variable is eliminated when the coalition combination is assembled. Public parameter count is $4(\lambda) + 6(p^{\text{DA}}) + 6(D) + 8(g) = 24$, but no 24-dimensional problem is ever solved.

The daily pacing target is receding: $D_i(t) = \text{clip}((D_i^{\text{day}} - h_2^{\text{inv}}) H / \text{steps left in day}, 0, D_i^{\text{max}})$; only step $k = 0$ is executed against realized price and renewable, then the window advances.

5 Online FACET clearing of the RTM GNE

Each agent's offline solution is an explicit piecewise-affine map: a set of *critical regions* (CRs), each a polytope $E_i \theta_i \leq f_i$ carrying an affine law $x_i = G_i \theta_i + g_i$. A *combination* $\mathcal{C} = (r_0, \dots, r_5)$ picks one

CR per agent; the equilibrium at combination $\mathcal{C}$ solves a block-linear system $M_x x = M_p \theta + M_1$. FACET finds and solves the correct combination online **without enumerating the $K^N \approx 9.7 \times 10^{20}$ combinations**.

The FACET method (what runs every 15-minute step)

1. **Per-agent point-location (warm-started).** Each agent locates its own CR at $[s_i; \theta]$ using the *previous* step's CR as a hint: test that CR and its facet-neighbours first (≈ 50 checks); only on a miss does a vectorized scan over that agent's own CRs run. This is per-agent (thousands of CRs), never over combinations.
 2. **Self-consistent coalition combination.** Assemble the located CRs into a combination; re-locate each agent given the others' aggregate and iterate to a fixed point — a genuine GNE (each agent best-responding to the shared aggregate).
 3. **Variational-GNE solve.** Solve $M_x x = M_p \theta + M_1$ by the *potential-minimizing* selection: when the coupling binds for ≥ 2 agents M_x is rank-deficient (a manifold of equilibria); we pick the one minimizing the coalition potential $\sum_i J_i$ = the variational GNE = social optimum (matches the centralized oracle). A plain solve would fail on that singular system.
 4. **Facet-neighbour walk (backup).** If the located combination is not yet a strict equilibrium, walk the combination's facet-neighbour graph (1st/2nd/3rd neighbour) for the closest valid one; a 1-hop min-potential refinement locks the variational combination.
 5. **ADMM fallback (rare; the only iteration).** If steps 1–4 fail — only at genuinely degenerate ceiling vertices — fall back to a warm-started distributed ADMM solve. Its result is used only if it converges; a non-converged result never becomes a warm start.
- All of steps 1–4 are *local computation on the onboard maps*: the inter-agent exchange is **exactly one decision broadcast per agent per step**, regardless of how many local checks run.

Privacy model (no centralized optimizer at either stage)

Agents share **only their bidding decisions, never their cost models** — in both the DAM (§3) and the RTM (§5). Consequently no party can solve the joint problem at runtime. A centralized QP is used **only offline** to certify accuracy (map = centralized) and to bound an iterative solver's round count; it is never in the loop.

5.1 The multiparametric maps (built once, offline)

Each agent's RTM mpQP is solved *once, offline* in its private parameter space θ_i by a combinatorial multiparametric-QP solver (PPOPT). The result is the explicit piecewise-affine map: a set of critical regions (CRs), each a polytope $E_i \theta_i \leq f_i$ carrying an affine control law $x_i = G_i \theta_i + g_i$. Only **4 distinct solves** are needed — the two PV agents and the two wind agents are identical — then expanded into the six agent slots. Crucially, the private θ_i box spans the *full* operating range of every parameter — price $\lambda \in [-50, 300]$, day-ahead anchor $p_i^{\text{DA}} \in [0, P_{\max, i}]$, demand $D_i \in [0, D_i^{\max}]$, renewable g — so the *same* maps are valid for *every* day and *every* DAM anchor. **The RTM maps are therefore built exactly once, never rebuilt**; only the day-ahead ADMM runs daily.

Agent	type	dim θ_i	decision x_i	CRs
PEM_Elec	grid	10	$[p_{0..3}] \in \mathbb{R}^4$	507
ALK	grid	10	$[p_{0..3}] \in \mathbb{R}^4$	507
PEM_PV, PEM_PV_2	renewable	14	$[p_{0..3}, cv_{0..3}] \in \mathbb{R}^8$	8359 (each)
PEM_Wind, PEM_Wind_2	renewable	14	$[p_{0..3}, cv_{0..3}] \in \mathbb{R}^8$	7335 (each)
Total		32,402 (4 distinct solves)		

Table 2: Offline multiparametric maps: per-agent critical-region counts. Each agent solves its own 10- or 14-dimensional mpQP; θ_i is compressed by the coupling “sum” variable so no 24-dimensional problem is ever solved. One-time offline cost ≈ 13 min; facet-neighbour graphs ≈ 1 s. **Generated once** and reused for both entire weeks.

6 Results — two 7-day runs (ERCOT 2025)

Both weeks use the full $[100, 900]$ band at both stages and the deployed FACET clearing (1 broadcast/step). “map = cent” is the max over the certified steps of $\|x_{\text{FACET}} - x_{\text{centralized}}\|_\infty$; “fallbacks” are steps that needed the warm-started ADMM escape; curtailment is renewable generation that could not be absorbed.

Day	λ_{RT} [\$/MWh]	map = cent [kW]	ADMM fallbacks	H ₂ met	curtail [kWh]
04-01	10–73	2.6×10^{-4}	0	167%	748
04-02	17–282	2.1	1	122%	342
04-03	11–31	6.9×10^{-6}	0	134%	275
04-04	9–60	6.4×10^{-6}	1	139%	300
04-05	11–23	5.9	0	163%	716
04-06	−1–234	5.5	0	131%	765
04-07	0–3553	3.4	1	126%	236
Week 1 (1–7 Apr)		≤ 5.9	3/672	140%	3381

Table 3: **Week 1.** 3 fallbacks / 672 steps (99.6% iteration-free). Interior days match the centralized equilibrium to $\sim 10^{-6}$ – 10^{-4} kW; boundary days (ceiling-bound) sit within ≤ 5.9 kW ($< 1\%$ of 900) of the social optimum — the multi-equilibrium residual of §7. Weekly curtailment 3381/43380 kWh = 7.8% of available renewable.

Day	λ_{RT} [\$/MWh]	map = cent [kW]	ADMM fallbacks	H ₂ met	curtail [kWh]
07-07	22–95	1.9	1	135%	165
07-08	21–151	4.2	1	132%	102
07-09	17–98	4.9	1	148%	181
07-10	11–79	2.6	4	156%	448
07-11	12–1754	1.6	1	163%	803
07-12	26–312	0.32	1	132%	434
07-13	16–82	3.2	0	129%	256
Week 2 (7–13 Jul)		≤ 4.9	9/672	142%	2390

Table 4: **Week 2 (summer, unseen)**. 9 fallbacks / 672 (98.7% iteration-free), including a \$1754 scarcity spike (07-11). Weekly curtailment 2390/34540 kWh = 6.9%.

Agent	Week 1 (April)		Week 2 (July)	
	H ₂ / target [kg]	curtail / avail [kWh]	H ₂ / target [kg]	curtail / avail [kWh]
PEM_Elec	608/462 (132%)	— (grid)	620/462 (134%)	— (grid)
ALK	418/333 (126%)	— (grid)	408/333 (123%)	— (grid)
PEM_PV (each)	335/231 (145%)	372/3521	358/231 (155%)	499/6043
PEM_Wind (each)	681/462 (147%)	1318/18169	679/462 (147%)	696/11227

Table 5: Per-agent weekly H₂ and curtailment. Every agent meets its contract (the receding $D_i(t)$ pacing keeps all on or above target); over-production ($>100\%$) reflects the day-ahead buying cheap power at the ceiling when $\lambda < a_i$. Renewable curtailment stays $\leq 8\%$ of available generation: the coalition lets renewable-rich agents absorb their own output rather than being forced to curtail. Rows “each” report one agent of the identical pair.

6.1 Critical-region and combination usage over the 2-week run

The number of *combinations* (one CR per agent) is $K^N = 507^2 \cdot 8359^2 \cdot 7335^2 \approx 9.7 \times 10^{20}$ — impossible to enumerate; FACET never does. Over the 1344 real-time steps of both weeks, FACET selected only **619 distinct combinations** — a fraction 6.4×10^{-19} of the possible set — and operation concentrates on a handful: the single most-used combination is selected on 117 of 1344 steps. This empirical concentration (~ 50 – 90 distinct combinations per day out of 10^{21}) is exactly what FACET’s neighbour-graph walk exploits, and why the 32,402-CR maps of Table 2 never need enumerating.

	Week 1 (April)	Week 2 (July)	Both weeks
Distinct combinations per day	48–67	53–86	—
Distinct combinations over the week	322	394	—
Distinct combinations used (union)			619
Of $K^N \approx 9.7 \times 10^{20}$ possible			6.4×10^{-19}
Real-time steps	672	672	1344
Most-used combination (visits)			117 / 1344

Table 6: Combination usage. Of $\sim 10^{21}$ possible combinations, real operation lives on **619** recurring ones over two weeks — the maps (Table 2) are built once and the FACET walk visits only this tiny recurring set.

Communication / data transfer. Every map step exchanges exactly **one decision broadcast per agent** (the FACET location, walk and refinement are local). Only the rare ADMM-fallback steps carry iterative rounds. Over each week the map steps contribute 1 round/step; the 3 (April) / 9 (July) fallback steps contribute the remaining rounds. An iterative GNE seeker run on *every* step would instead need $\sim 10^2$ – 10^3 agent broadcasts per step.

6.2 Figures

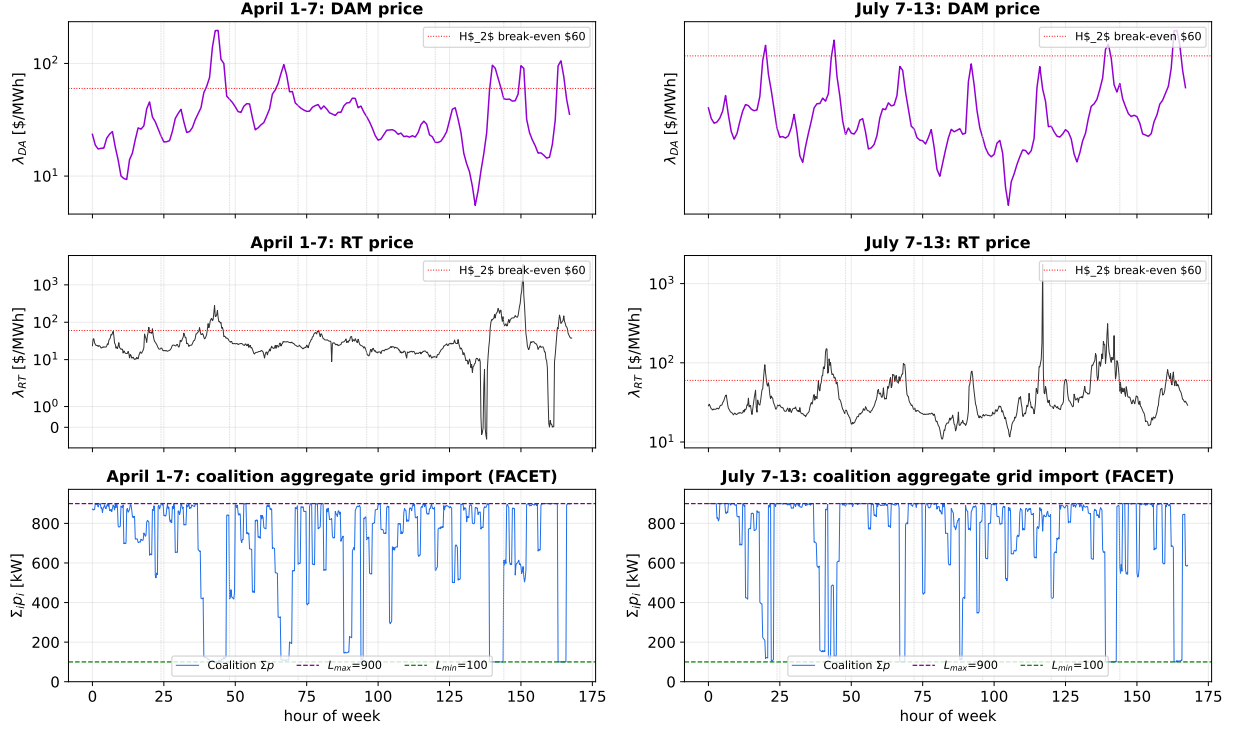


Figure 1: Coalition aggregate grid import $\Sigma_i p_i$ (bottom) against the RT price (top, symlog) over each week, from the FACET clearing. The aggregate stays within $[L_{\min}, L_{\max}] = [100, 900]$ (dashed) at all times; it rides the ceiling on cheap hours (buy H₂) and retreats when the RT price exceeds the H₂ break-even.

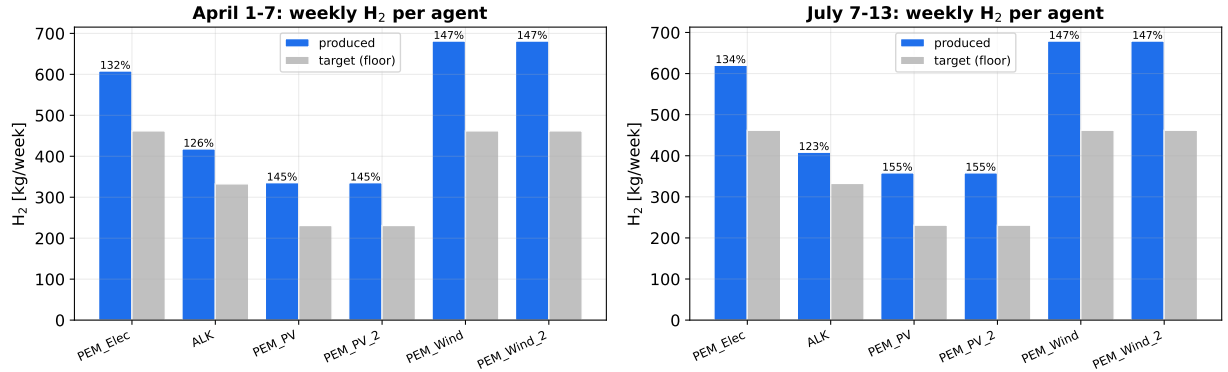


Figure 2: Weekly H₂ per agent, produced vs. the contract floor (target). Every agent clears its floor; over-production reflects buying cheap power at the ceiling. Labels show % of target.

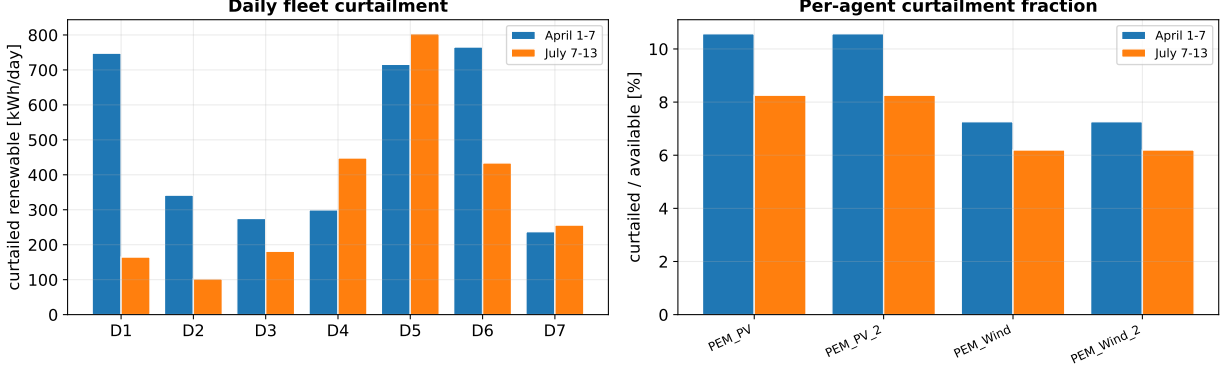


Figure 3: Left: daily fleet renewable curtailment (kWh) for both weeks. Right: per-agent curtailment as a fraction of available renewable — $\leq 8\%$: the coalition lets renewable-rich agents absorb their own output rather than being forced to curtail.

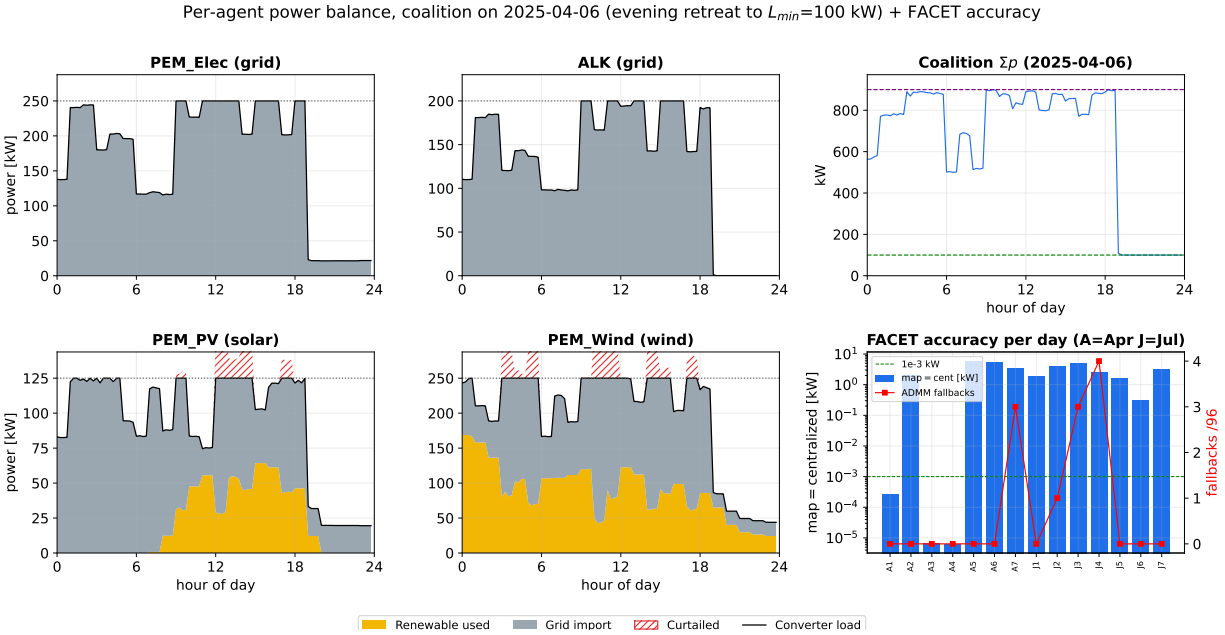


Figure 4: Per-agent power balance on 2025-04-06 (coalition) — a day the RT price swings from $-\$1$ to $\$234/\text{MWh}$, so the coalition both rides the ceiling (cheap/negative hours) *and* **retreats to $L_{min} = 100$ kW when expensive**: the renewable agents opt out of the market (grid import $\rightarrow 0$, running on their own generation, gold) while the grid agents cover the floor (top-right aggregate touches 100). The converter load (black) = renewable used (gold) + grid import (grey); curtailed renewable is the red hatch above the load. Bottom-right: FACET accuracy (map = centralized, log, bars) and ADMM fallbacks (red) per day across both weeks — interior days $\sim 10^{-6}$ kW, ceiling-bound days a few kW, fallbacks $\leq 4/96$.

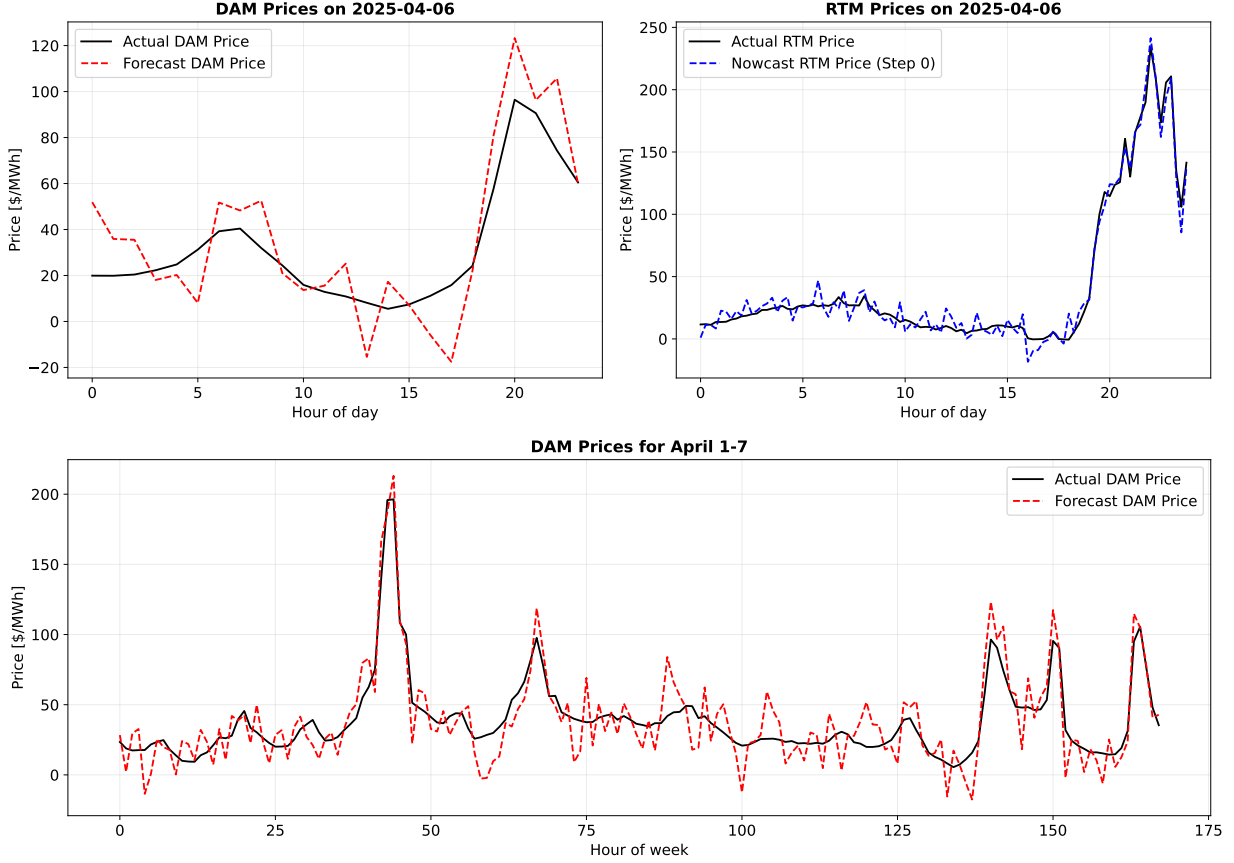


Figure 5: Actual vs. forecasted prices for the DAM (top left) and RTM (top right) on a representative day (2025-04-06), and the DAM prices for the entire week (bottom). The DAM forecast is evaluated once, a day ahead; the RTM nowcast is re-evaluated at every 15-minute receding step with fresh noise — illustrating the forecast-vs-realized split the RTM must correct for online.

7 Limitations (honest)

Scope and residuals

- **Multi-equilibrium residual at the ceiling.** With the full $[100, 900]$ band the day-ahead schedule buys cheap power up to the 900 kW ceiling, so real-time dispatch frequently binds the coupling, where the GNE is *non-unique* (Hall & Bemporad 2025, §II-C: rank-deficient M_x , overlapping CRs). FACET’s potential-minimizing solve targets the variational GNE, but the 1-hop refinement does not always reach it, so on ceiling-bound steps the online solution can sit a *few kW* ($< 1\%$ of 900) from the social optimum, and $\sim 0.9\%$ of steps use the warm-started ADMM fallback. *Planned fix (deferred)*: apply the paper’s common-multiplier (homogeneous-multiplier) v-GNE selection *online at the located boundary combination* (their eqs. 16–17), which selects the variational branch exactly and would drive

these steps to $\sim 10^{-4}$ kW and remove most fallbacks. The offline machinery exists (`gne_combiner.filter_variational_kkt`).

- **H₂ over-production.** The daily H₂ target is a *floor*; with cheap ERCOT power and \$3/kg H₂, buying to the ceiling is profitable, so the fleet over-produces to $\sim 140\%$. Economically rational, but a conscious design point.
- **Validation oracle.** The centralized QP is used only offline to certify accuracy; agents never share models, so it cannot run in deployment.
- **Model scope.** No batteries, no 65-min binary participation gate, no export — kept out to remain a convex (mp)QP at both stages. The semi-continuous $\{0\} \cup [L_{\min}, L_{\max}]$ market-exit rule (non-convex) is out of scope for the explicit-map method.

8 Reproduction

`simple_game/rhg_mpqp.py` (per-agent $H=4$ private- θ mpQP; holds the locked spec $L_{\min}/L_{\max} = 100/900$, fleet); `rhg_offline.py` (one-time PPOPT solve $\rightarrow$ `out/rhg_agent_sols.pkl`); `dam.py` (distributed day-ahead ADMM + centralized oracle); `rhg_online.py::solve_step` (FACET point-location + v-GNE solve + neighbour walk + warm-started ADMM fallback + data-transfer accounting); `rhg_week.py` (distributed DAM + receding closed loop, curtailment tracking); `rhg_figs.py` (figures). Data in `data/ercot/`; locked spec in `FORMULATION.md`; status/handoff in `STATUS_dam_rtm_boundary.md`.